SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

BSM 451
INTERNET OF THINGS AND APPLICATIONS

Machine-to-Machine Communication, M2M

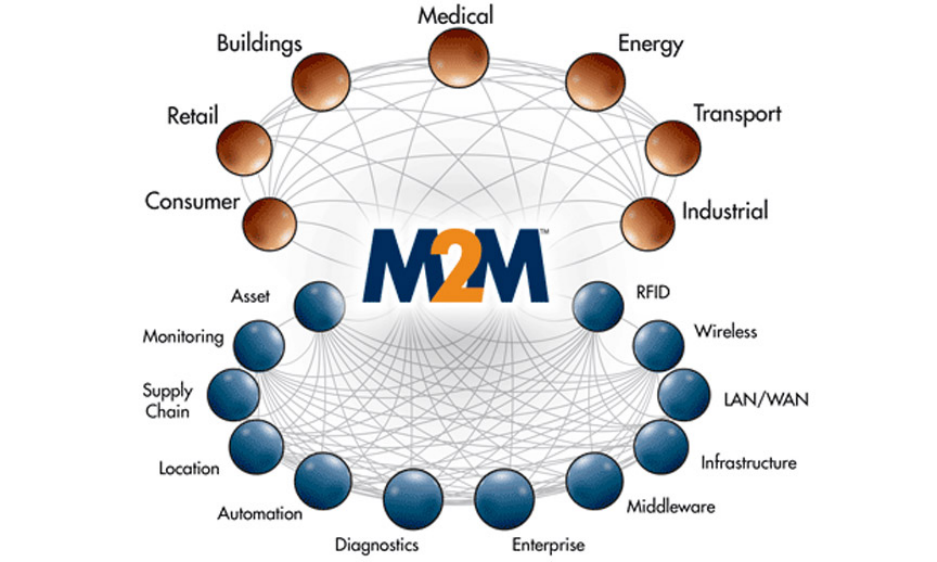
Assoc. Prof. Cüneyt BAYILMIŞ
Researcher Dr. Ünal ÇAVUŞOĞLU

What is M2M?



Introduction

- One of the major research area in the field of information and communication Technologies is M2M.
- M2M is considered a part of IoT ecosystem.
- M2M is end-to-end communication of devices (machines) without human action (or as less as possible).
- M2M applications are used in sectors such as transportation, health, security, etc.



Machine-to-Machine Communication

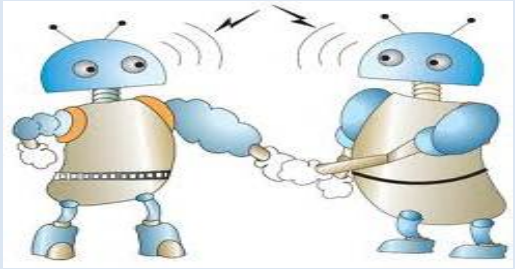


- M2M is communication of two or more devices under pre-determined rules and protocols through wired or wireless medium.
- OECD: M2M devices are actively communicating using wired and wireless networks, are not computers in the traditional sense and are using the Internet in some form or another
- ITU: M2M is communication between two or more devices with no or minimum human intervention.
- ETSI: M2M is communication between two or more devices with no human intervention
- CEPT: M2M is type of mostly or all automatic communication over an electrical communication network between two or more information and communication technology (ICT) component.

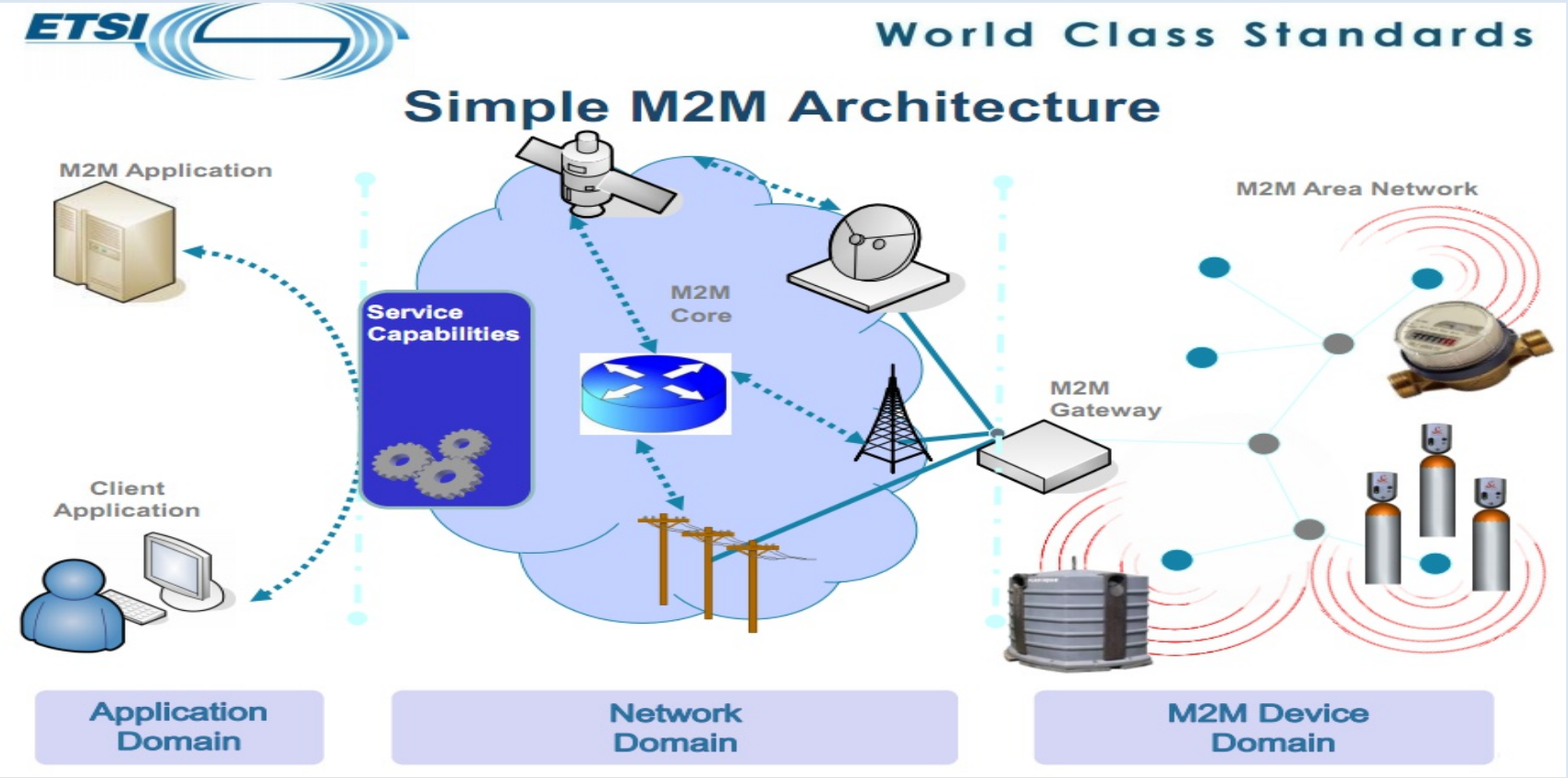
What are the benefits of M2M?

- Reduced cost,
- Increased efficiency,
- New business models,
- Improved quality,
- Prediction of incidents, e.g., accidents



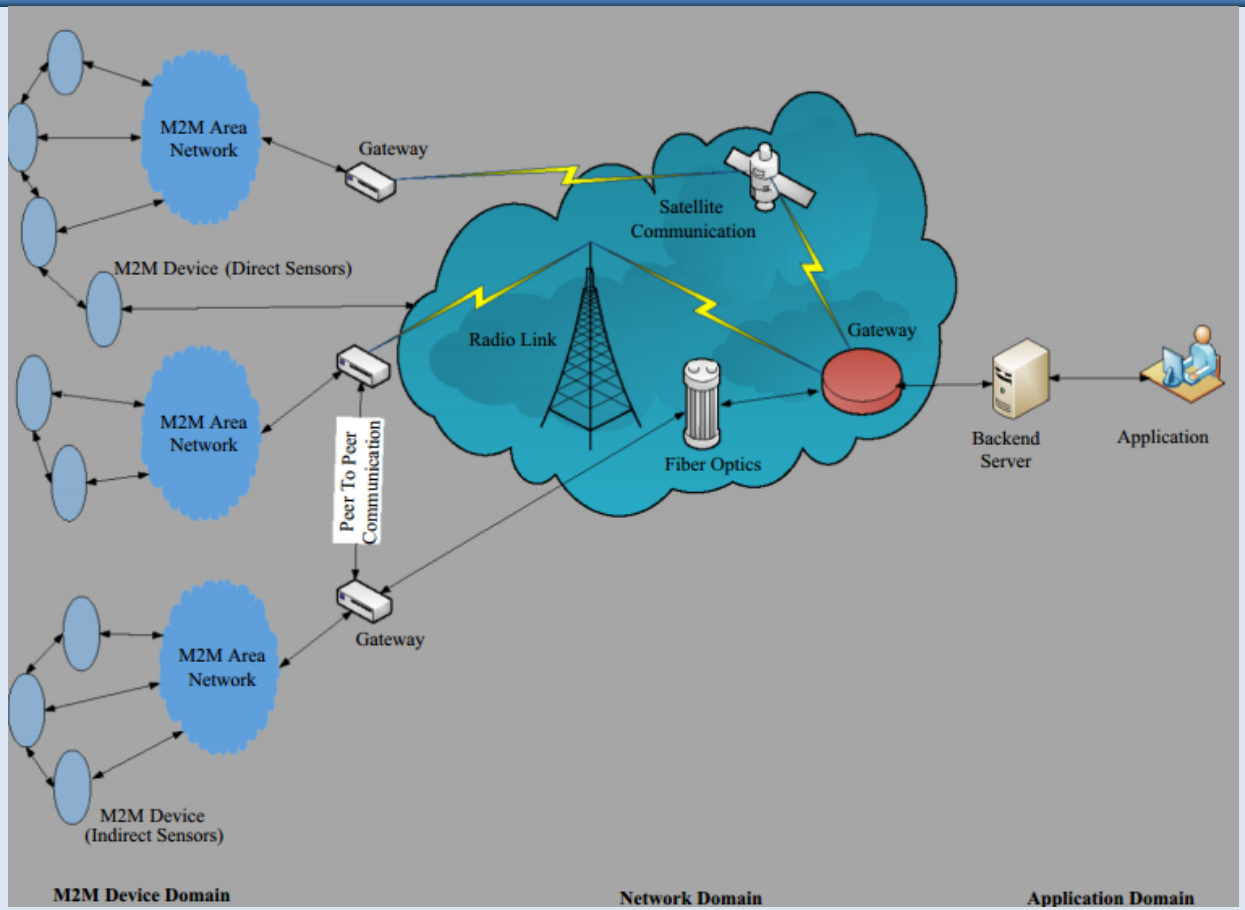


M2M Architecture



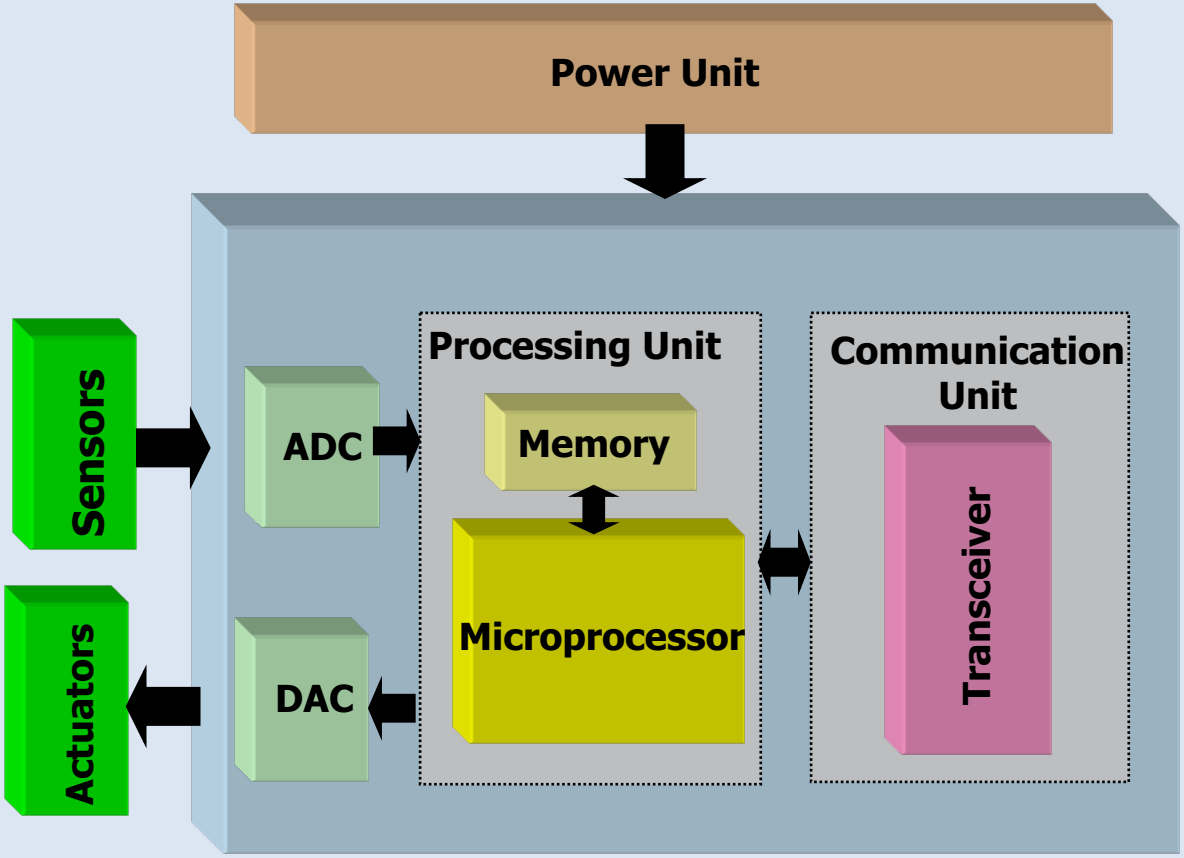
- Data gathering triggered by autonomously by M2M devices or by user interaction
- Transferring of digitally transformed data obtained from application domain to other machines
- Domain of other M2M devices. This is the place where the main operation is performed for the requested service.

M2M Communication System Architecture



- M2M network components
  - PSTN,
  - Wired networks,
  - In-building power cables,
  - WiFi,
  - 2G/3G/4G/4.5G/5G
  - Satellite,

M2M Device


Power Unit
Sensors
Actuators
ADC
DAC
Processing Unit
Memory
Microprocessor
Communication Unit
Transceiver

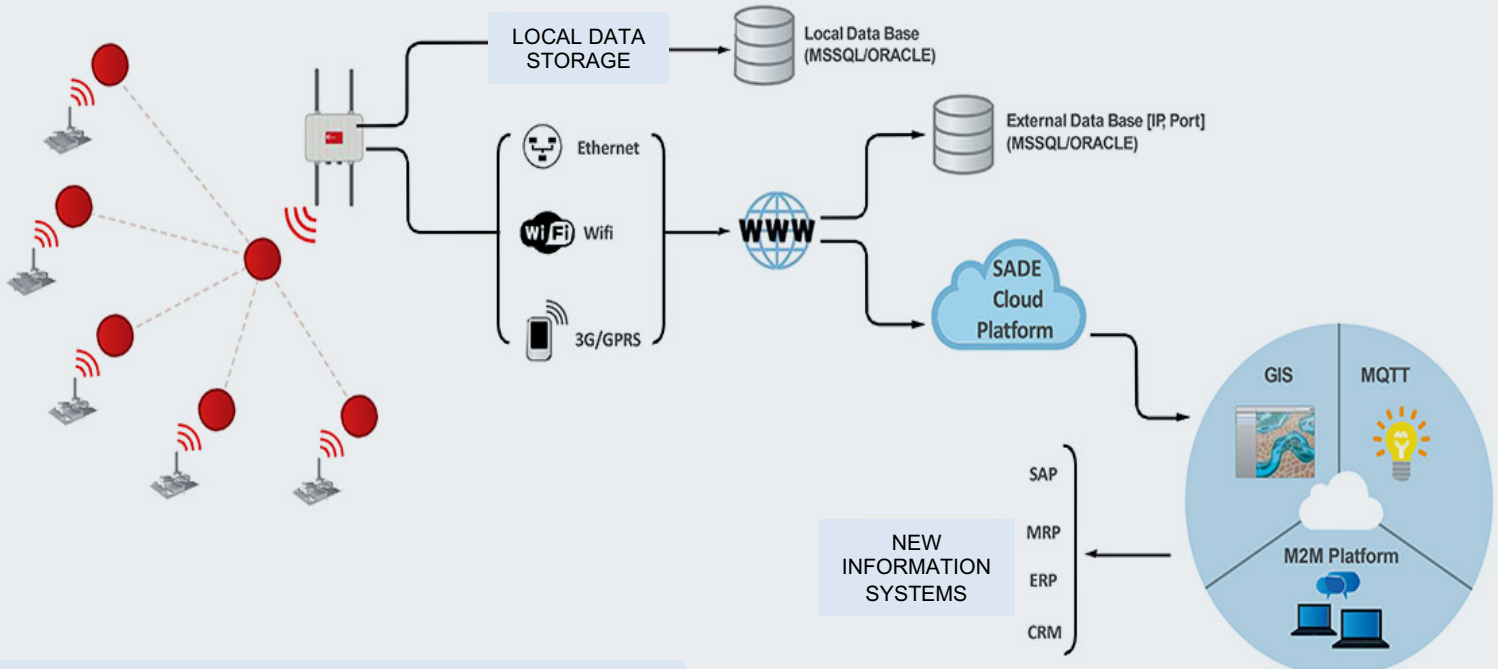
M2M Basic Components

- **Data Gathering Technologies:**
  - Sensors, RFID, BLE Beacon



- **Communication Technologies:**
  - Ethernet, WiFi, Bluetooth, ZigBee, Z-Wave, GSM, GPRS, 3G / 4G / 4.5G

How does M2M work?

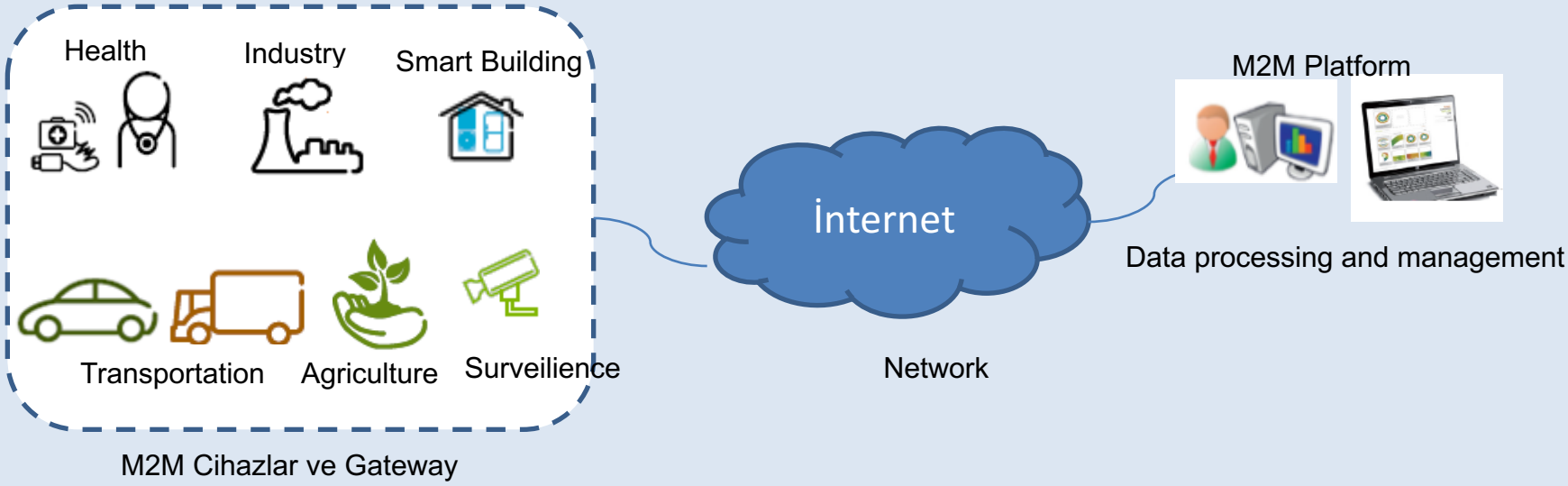


Sample diagram for data gathering with wireless sensors

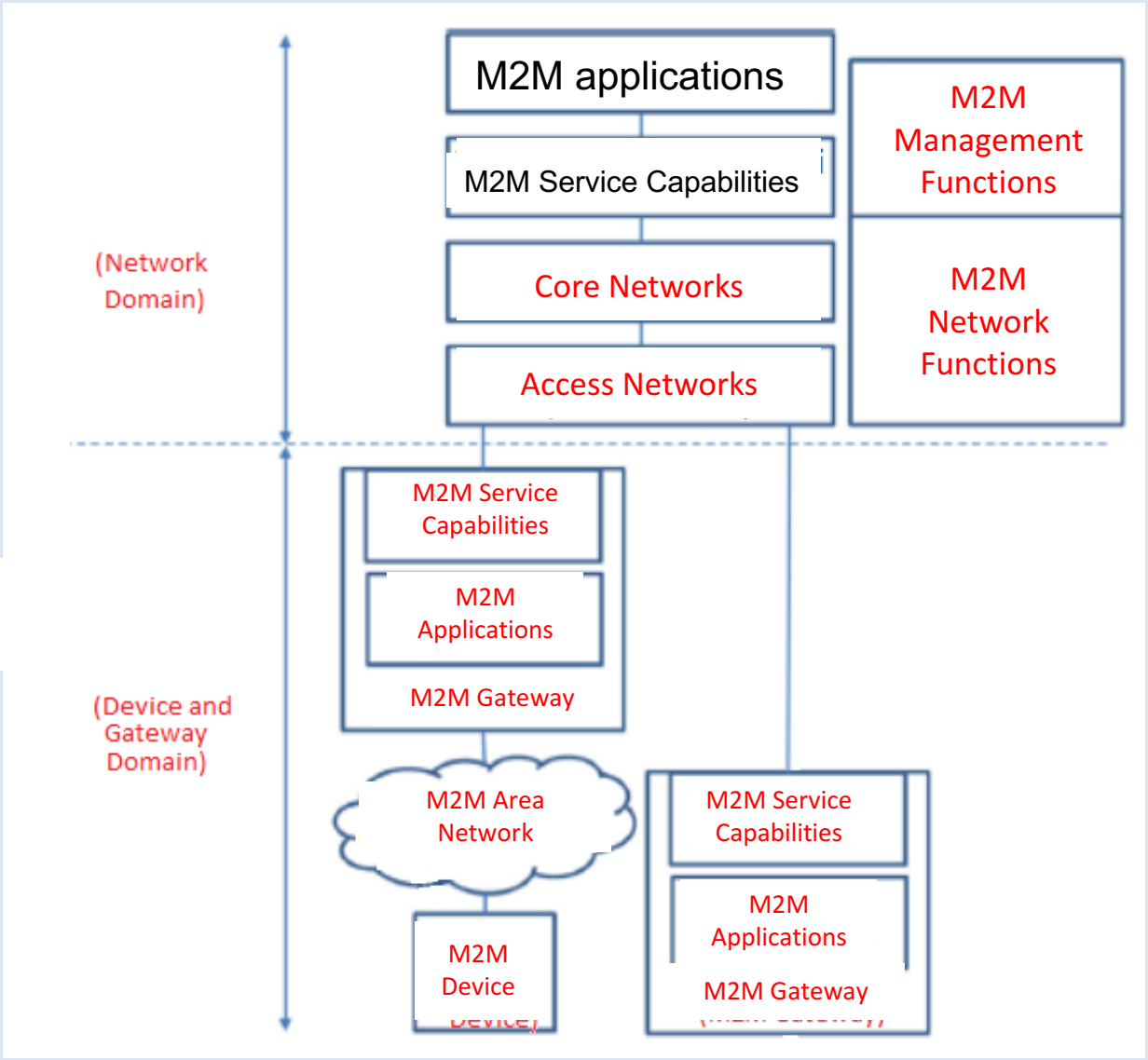
- What are the benefits of M2M Technology?
  - Configuration and management of M2M devices in the field,
  - Data gathering from M2M devices in the field and data analysis,
  - Sending tasks to M2M,
  - Cost efficiency to M2M users.

M2M

- General characteristics of M2M systems:
  - Many node/device connection support,
  - Interoperability,
  - Autonomous,
  - Self organizing,
  - Power efficiency,
  - Reliability/Security,



M2M Functional Architecture



M2M Functional Architecture

Device and Gateway Area

M2M Device

- M2M device has M2M applications and is service capable.
- Basic types and advanced types.

M2M Area Network

- Area network that connects between M2M devices and M2M network gateways.
- Local and personal network area.
- Includes technologies such as Bluetooth, ZigBee, KNX, etc.

M2M Gateway

- It provides connection of M2M devices in M2M area network to internet. It provides neccessary tranformation mechanisms.
- Network Gateway has also M2M applications and service capabilities.

M2M Functional Architecture

Network Area

- Access Network
  - Allows devices to communicate with core network.
  - xDSL, WLAN, WiMAX, Satellite.
- Core Network
  - IP connection, service and network control, connection with other networks.
- M2M Service Capabilities
  - Under core network functions.
  - Abstracting network functions to simplfy applications.
- M2M Applications
  - Capable of using M2M service capabilities with open interfaces.
- Network Management Functions
  - Functions to manage configurations of access and core networks, error management.
  - Abstracting network functions to simplfy applications.
- M2M Management Functions
  - Neccessary functions to manage M2M service capabilities in network area.
  - Functions for security.

M2M Data Traffic

Since M2M applications can be used in various goals in various fields of power to health, traffic types depend on the application area.

- elastic applications,
- real-time applications,
- rate-adaptive applications,
- mobile streaming,
- smart grid,
- emergency services,
- e-healthcare,
- information and navigation services,

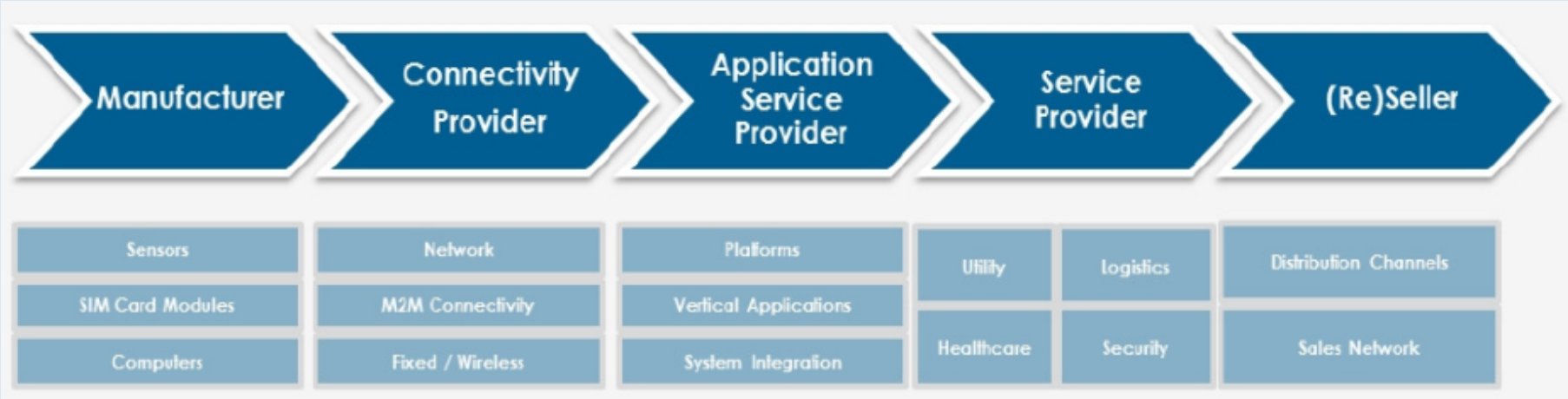
M2M Service Capabilities

- Application authorization,
- General communication,
- Reachability and addressing,
- Security,
- Operation management,
- Interconnection

How to select M2M network technology

- Coverage,
- Number of supported devices,
- Power consumption,
- Ease of setup and maintanence
- Open or closed system,
- Flexibility in user interactions,
- Application Type that is supported,
- Mobility,
- Secuirty and privacy,
- Demo support,

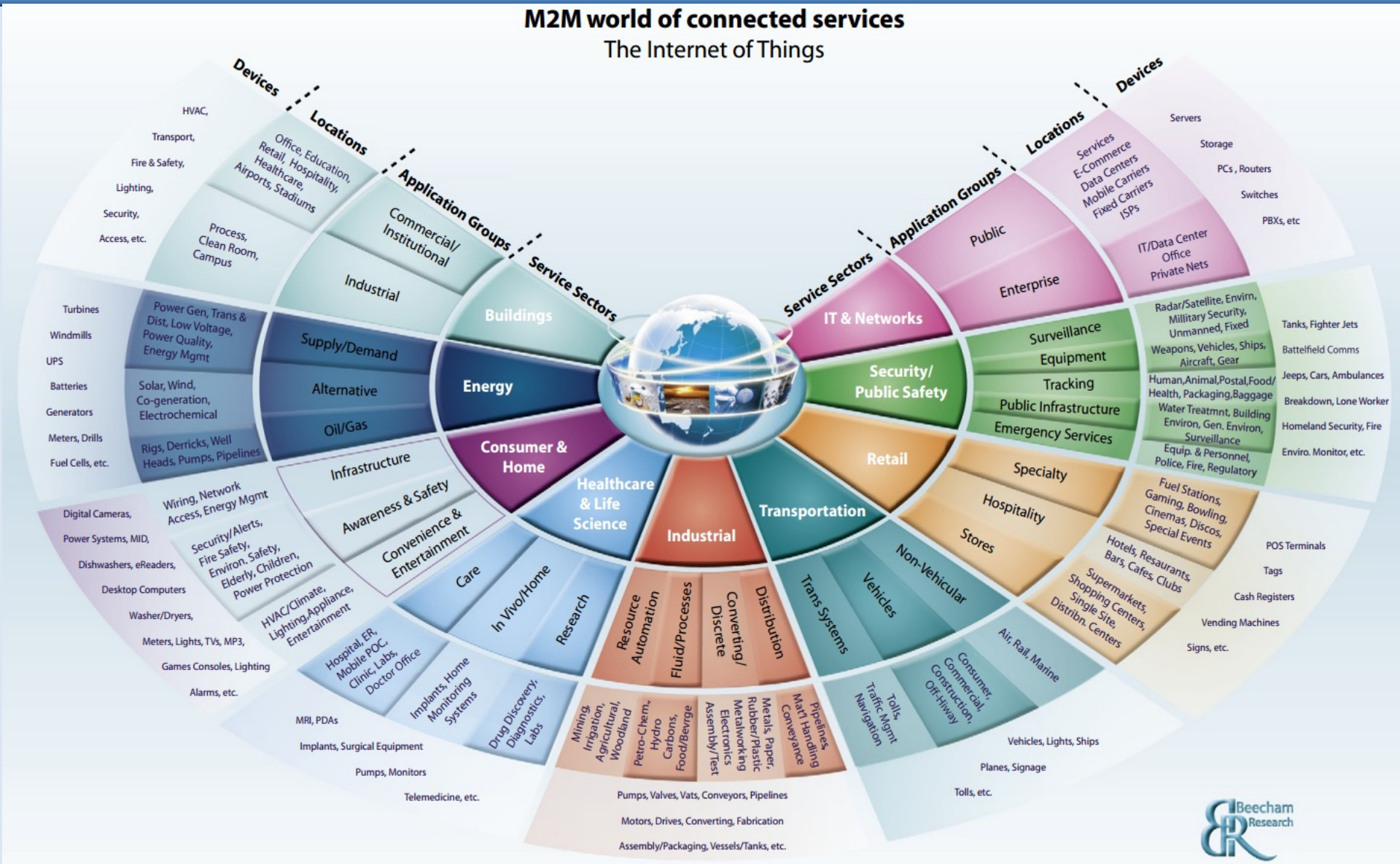
M2M Value Chain



- **Where would you be in this chain as a computer engineer?**

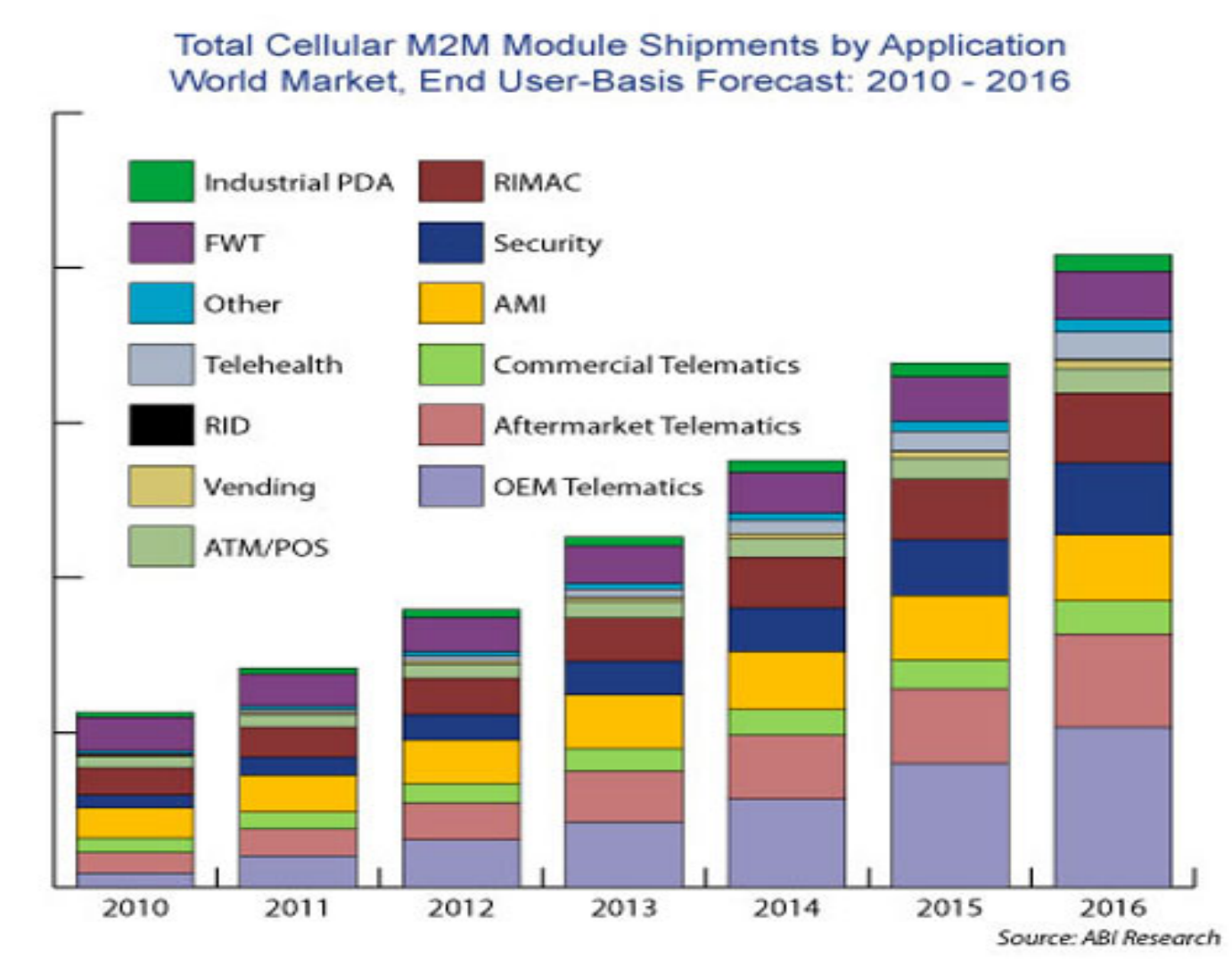


M2M Application Area

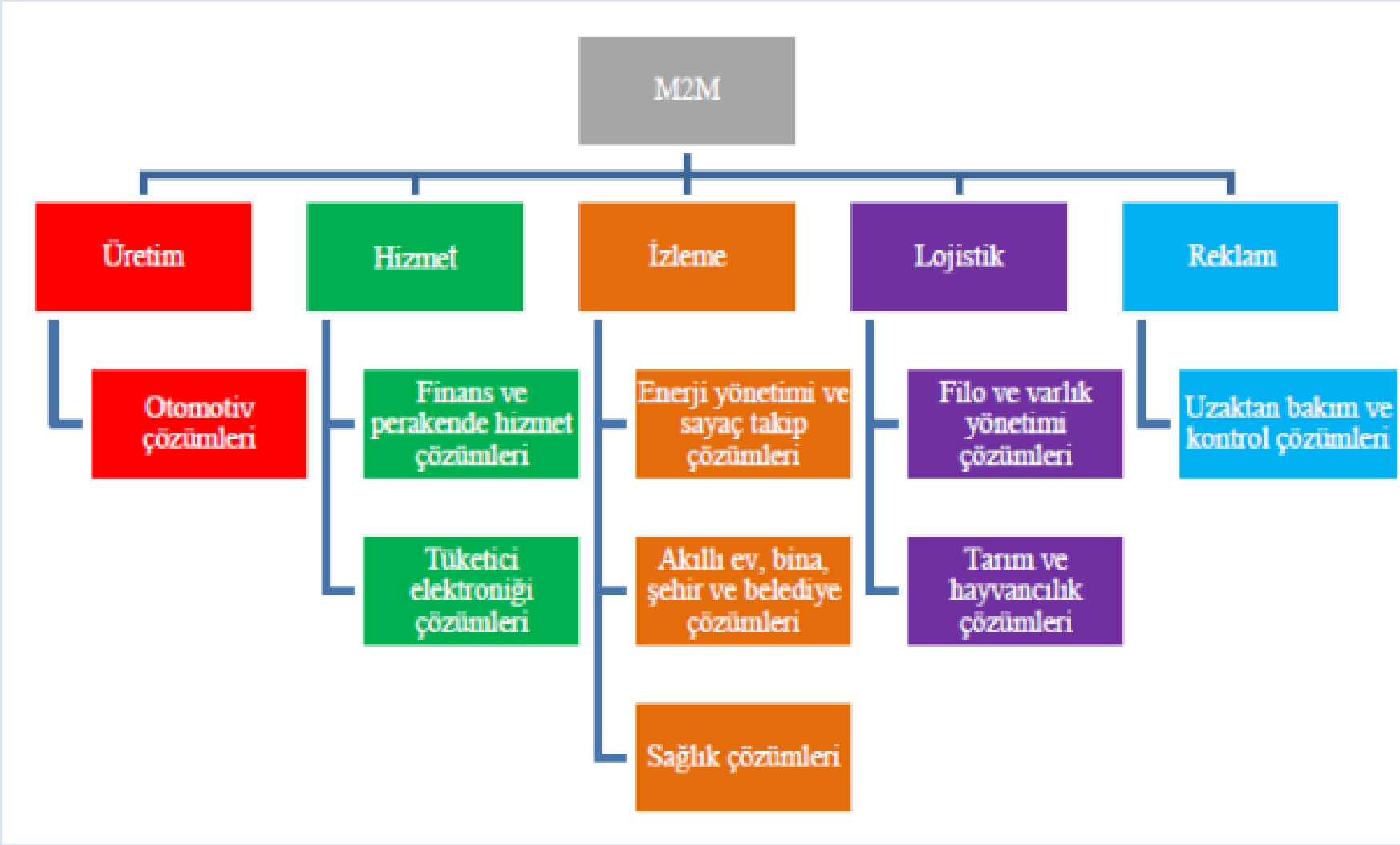


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Application Areas and Market in the World

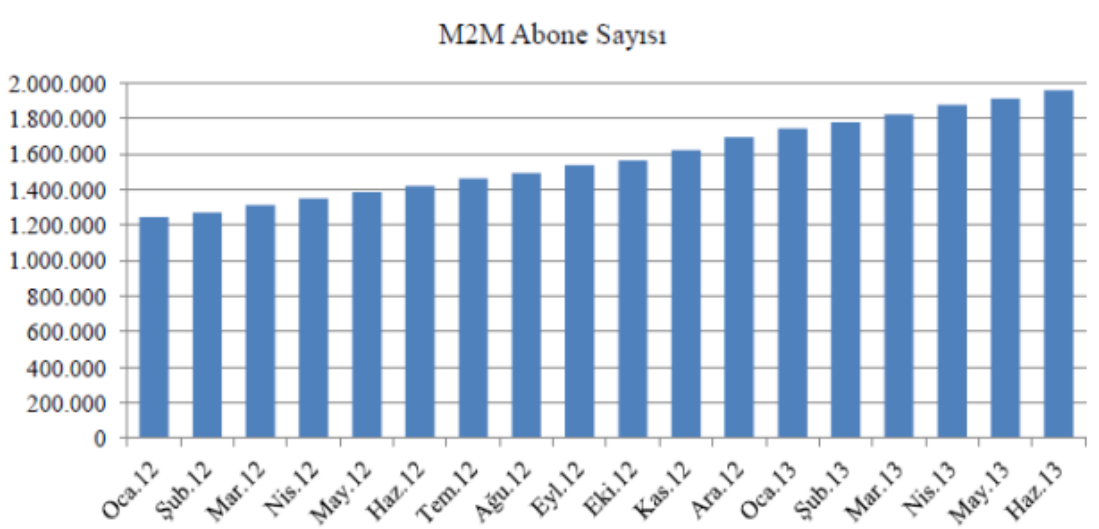


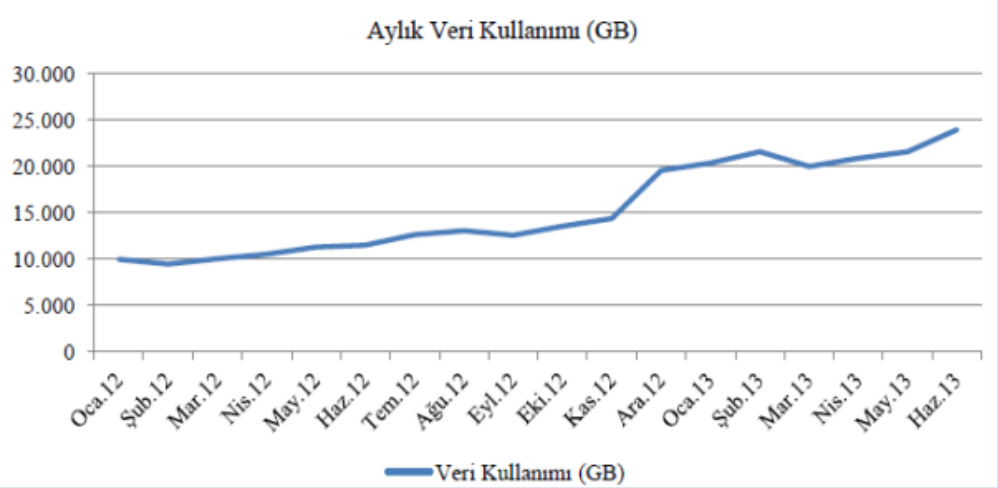
Application Areas and Market in the Turkey

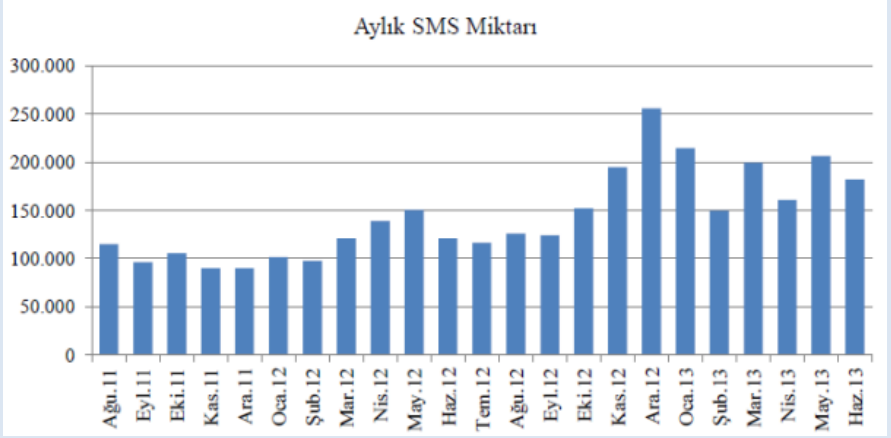


Source: BTK M2M Raporu, 2013

Application Areas and Market in the Turkey







Source: BTK M2M Raporu, 2013

M2M vs. IoT

	M2M	IoT
Application and Services	- Solution based on targeted problem - Single application, single device - Communication and Device Centric	- Based on innovation - Multi-application, multi-device - Information and Service Centric
Business	- Closed to business operations - B2B - In-firm located - Established value chain	- Open market placement - B2B, B2C - Cloud located - Improving ecosystem
Technology	- Specialized device solutions - De Facto and certified - Specific closed data formats and service definitions - Closed personalized software development	- General commercial devices - Standards and open source - Open APIs and data definition - Open software development

Practice

- Describe M2M briefly?
- Explain M2M architecture?
- What are the differences between M2M and IoT?
- What are the components of ETSI functional architecture?

References

- "Makineler Arası İletişim (M2M)", Bilgi Teknolojileri ve İletişim Kurumu, 2013.
- J. Höller, V. Tsiatsis, C. Mulligan, S. Karnouskos, S. Avesand, D. Boyle, "Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", Elsevier, 2014.
- P. K. Verma et al., "Machine-to-Machine (M2M) communications: A survey", Journal of Network and Computer Applications, Vol. 66, 83-105, 2016.